Unit 1

Introduction

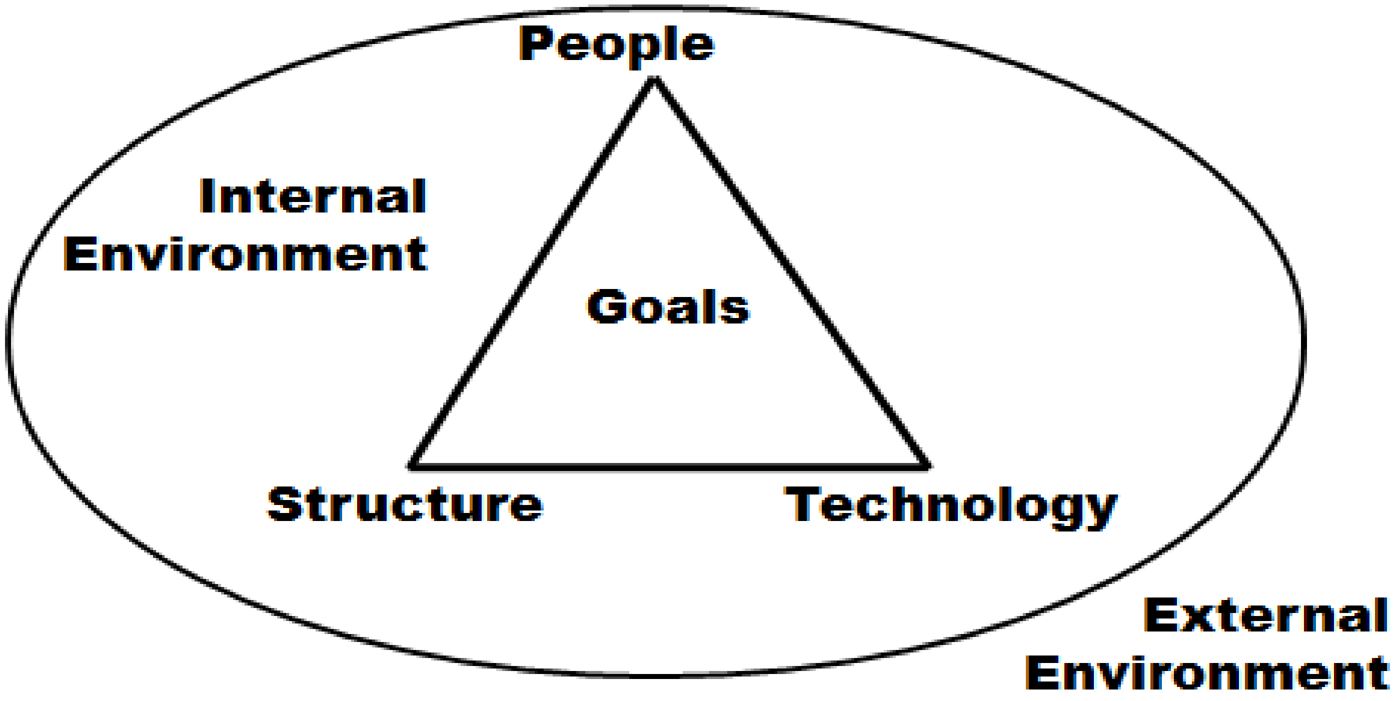
Organization

- It’s a Social Structure
- An entity in which two or more people work interdependently through structured patterns
- Coordination of man, machine and materials
- For a purpose of accomplishing a set of goals
    - Create value for its stakeholders, stockholders, employees, community, customers

Organization is a .....

- Goal Oriented System
  - People work together for specific goals
- Social System
  - People work in a group
- Technological System
  - People use the knowledge and techniques they possess

Organization in 1 picture



Why an Organization

- People work together to achieve certain goals
- When organized, they can have better performances
- Fulfill basic needs
- Its better to work in group than work alone
- Better resources, diverse views and knowledge, more experiences, great potentials

- The objective must be common for the organization
- Rules and Regulations must be organized, so no conflicts and confusion occur
- People related should be willing to cooperate with each other
- Have greater communications with each other

Organization in Society

- The goals the organizations set to achieve directly or indirectly affects the society
- Development in education system, healthcare, technologies have helped us live a better life
- Hospitals, Colleges and Universities, Banks, Telecommunication, Technologies, Entertainment industry

System approach of Organization

- Consists of interdependent parts that work together to continually monitor and transact with the external environment
- Important to study the change in the external world
- Changes in technologies, methods, customer needs, market trends, rules and regulations
- Feedback from the customers

MANAGEMENT

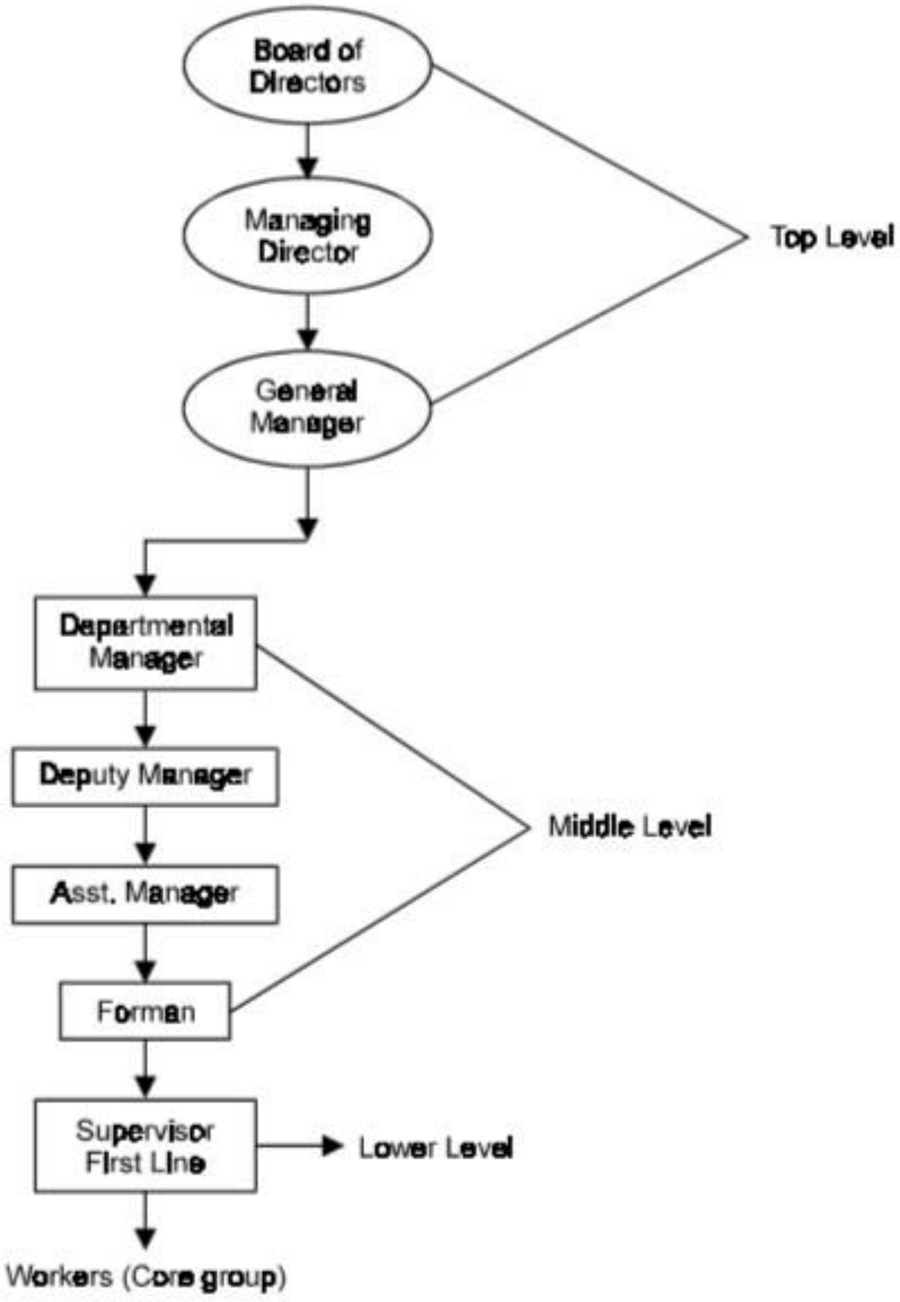
- In simple words, getting things done through other people
- In an organization, management is the act of getting people together to achieve the set goals using the available resources effectively and efficiently
- Brain of an Organization
- Coordination of human, material resources and technology
- Management's existence as old as the human origin
    - General sun Tzu, the art of war, 6th century BC
    - Chanakya's Arthashastra, 300 BC

Function/Importance

- Proper Planning, determine what is to be achieved
- Proper Organizing, allocate resources and establish the means to accomplish the plan
- Proper influence, motivate and lead personnel towards the goal
- Controlling activities in the organization, compare results achieved to the planned goals

Levels of Management

- Depends upon the size, complexity and the nature of organization
- Commonly this structure is followed in most of the organization →



- Top level Management
  - Usually resides in the Top Layer of the Pyramid.
  - Higher Authority
  - Sets the goals, objectives, policies, and budgeting of the organization
  - Main functions involve the decision making
  - Sets the rules and regulations, issues order to the lower levels, guidelines and instructions

- Middle Level Management
  - Usually resides in the Middle Layer of the Pyramid.
  - Head of the functional departments
  - Mediator between the top level management and the lower level
  - Main function is to implement the policies set by the top level

- Lower Level Management
  - Usually resides in the Lowest Layer of the pyramid
  - Responsible for the day to day activities
  - Supervision of the workers and labors
  - More involved in working to achieve the goals set by the organization

Function of Management

- Planning
  - Thinking before doing
- Organizing
  - Developing a framework that relates all personnel, work assignments and resources
- Directing
  - Requires leadership
  - Giving guidelines to the personnel related to the definite objectives
  - Supervising

- Coordinating
  - Required for the efficient and effective run of the organization
  - Managers are responsible for coordination and communication between different individuals working under the organization, to achieve the same set of goals
- Controlling
  - Watch the activities of the organization
  - Compare the results achieved with the goals set

SCOPE OF MANAGEMENT

1. Production Management
2. Marketing Management
3. Financial Management
4. Personnel Management

Production Management

Planning, organizing, directing and controlling the production function so as to produce the right goods, in right quantity, at the right time and at the right cost. Includes the following activities:

- designing the product
- location and layout of plant and building
- planning and control of factory operations
- operation of purchase and storage of materials
- repairs and maintenance
- inventory cost and quality control
- research and development etc.

Marketing Management

Identification of consumers needs and supplying them the goods and services which can satisfy these wants. Involves the following activities:

- marketing research to determine the needs and expectation of consumers
- planning and developing suitable products
- setting appropriate prices
- selecting the right channel of distribution, and
- promotional activities like advertising and salesmanship to communicate with the customers

Financial Management

Seeks to ensure the right amount and type of funds to business at the right time and at reasonable cost. Comprises the following activities:

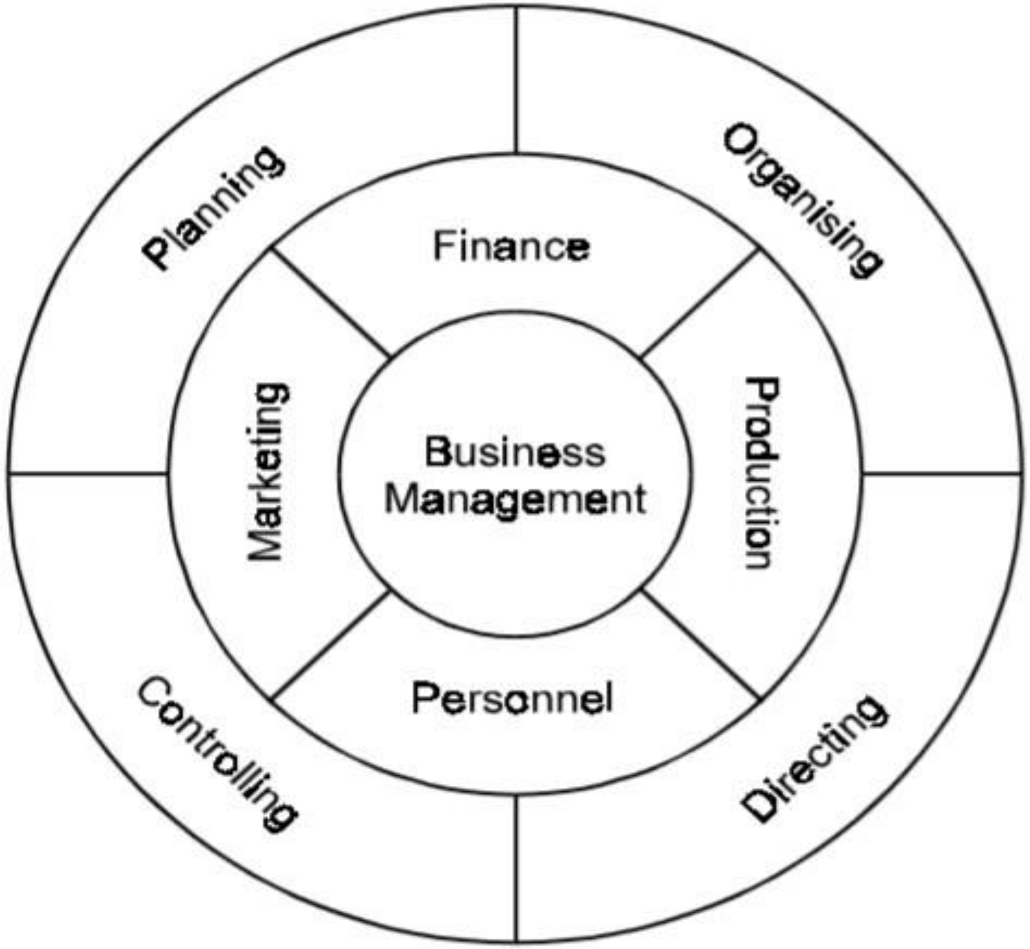
- estimating the volume of funds required for both long-term and short-term needs of business
- selecting the appropriate source of funds
- raising the required funds at the right time
- ensuring proper utilization and allocation of raised funds so as to maintain safety and liquidity of funds and the credit- worthiness and profitability of business, and
- administration of earnings Thus, financial management involves the planning, organizing and controlling of the financial resources.

Personnel Management

Involves planning, organizing and controlling human resources of an organization. It consists of the following activities:

- Manpower planning
- Recruitments,
- Selection,
- Training
- Appraisal,
- Promotions and transfers,
- compensation, employee welfare services, and personnel records and research

Scope of Management at a glance



Principles of Management

- Henry Fayol (1842-1925), French Mining engineer and Management Consultant.
- First person to analyze the functions of Management
- Made three major contribution to the theory of management
  - A clear distinction between technical and managerial skills
  - Identified functions constituting the management process
  - Developed principles of management
- Fayol described management as a scientific process built up of five elements
  - Planning
  - Organizing
  - Directing
  - Coordinating
  - Controlling

- Fayol's Principle
  Developed a set of 14 principles

  - Division of Labour
    - Work should be divided into individuals and groups as per their specialization. Work Specialization
  - Authority and Responsibility
    - Authority is the right to decide, to direct others, to take action, to perform certain duties in achieving the organizational goals
    - Responsibility is an obligations to perform work activities
  - Discipline
    - A successful organization requires the common effort of the workers
    - Penalties should be applied when necessary

- Line of Authority
  - A clear chain or authorities from top to bottom level of management
  - Scalar Chain
  - Authority should be defined at each level of Management and it should flow from top to bottom
- Centralization
  - Centralization is the degree to which authority rests at the top level
  - Fayol defined Centralization as lowering the importance of subordinates role, where as decentralization is increasing the importance
  - Whether determining centralization or decentralization depends upon the specific organization
- Unity of Direction
  - The entire organization should be moving towards a common objective in a common direction
- Unity of command
  - Employees should have only one boss
- Order
  - Each employee is put where they have the most value
- Initiative
  - Management should encourage employee initiatives, self direction

- Equity
  - Treat all employees fairly in justice and respect
- Remuneration
  - Fair rate should be determined for paying worker’s effort keeping in mind many variables like, cost of living, qualification, business conditions and the amount of responsibility
- Stability of Tenure
  - Long term employment is important for success
- General interest over individual interest
  - Importance should be given to the success of the organization rather than individual success
- Espirit de Corps
  - “Union is Strength”- refers to harmony and mutual understanding among the members of an organization

ORGANIZATION

- An organization is a social group which distributes tasks for a collective goal.
- Organization is the foundation upon which the whole structure of management is built.
- It is the backbone of management Organizations are human associations.
- There are a variety of legal types of organizations, including: corporations, governments, non-governmental organizations, international organizations, armed forces, charities, not-for-profit corporations, partnerships, cooperatives, and universities.

CHARACTERISTICS OF ORGANIZATION

- There should be two or more people and perform various function
- Organizations are goal oriented. They are created to achieve common goals.
- Organizations have continuity
- Organizations use technology to transform inputs into out puts
- Organizations have structures
- Organizations are open system
- Organizations are of many types: business, government, service, unions, international, political, cultural, etc.
- Organizations have several level- top, middle, lower as well as differentiated functions
- An Organization is managed by its leader

Types of Organization

1. Formal Organization

2. Informal Organization

3. Virtual Organization

Formal Organization

- Formal Organization has a structure that is consciously designed to enable the people of the organization to work together for accomplishing common objectives.
- Thus, formal Organization is more or less arbitrary structure to which the individual must adjust.
- It tells individual members to do certain things in specific manner to obey orders from designated individuals and to cooperate with others.
- Coordination also proceeds according to a prescribed pattern in the formal Organization structure.

Formal Organization is built on certain principles, i.e. around four key pillars, namely,

(a) division of labor, i.e. the whole work is divided into a number of small operations and each operation is performed by a different person so that there is maximum specialization

(b) scalar and functional processes, which implies that the Organization grows both vertically and horizontally,

(c) structure, which refers to the overall arrangement ensuring proper balance between different parts of the organization and secures the execution of all operations and the achievement of organizational objectives

(d) span of control, refers to the number of subordinates directly reporting and accountable to one superior.

Characteristics of formal organization

- It is deliberately created by management
- It is created to accomplish predetermined goals
- It is based on division of work and job specialization
- Authority-responsibility of every position is clearly defined
- Communication channel is through scalar chain
- Members are guided by formal policies, plans, rules and procedures.
- It has long life in terms of continuity
- Much emphasis is placed on efficiency, discipline, conformity, consistency and control
- It is slow in adapting in environmental changes.

Criticisms of Formal Organization:

- individuals are ignored in determining the interactions, communication and accountability;
- the ideal relationship assumes that rational human beings will stick to rules and regulations but such assumption is hard to find in reality;
- Assumption that punishment or reward always brings a reaction among humans (rabble hypothesis) also fails to succeed as humans are not always motivated by rewards and punishment.;
- Rules and regulations of a formal Organization is too rigid and not sensitive to changing times and circumstances, thus becomes difficult over time to achieve the goals of the organization.

Informal Organization

- Informal organizations refers to the relationship between people in the organization based on personal attitudes, emotions, prejudices, likes, dislikes, etc.
- These relations are not developed according to procedures and regulations laid down in the formal organization structure; generally large formal groups give rise to small informal groups.
- These groups are not preplanned; rather develop automatically/spontaneously according to the organizational environment.

- Informal structure created within an organization
- Based on the factors like same language, culture, languages, personal attitudes, same tastes or any other factors
- Informal organization exerts a strong influence on employee behaviors, if management can understand its nature, it can contribute to organization's productivity
- Are formed informally within the organization, by socializing and interpersonal relationships within the individuals over the breaks, or other social functions
- Not necessarily formed to accomplish the organization goals, rather formed to fulfill the social activities
- An informal leader is chosen within the informal organization who can have greater power than the formal leader over getting things done in the organization

Characteristics of informal organization

- It is unplanned and spontaneous.
- It is based on common interest, attitude and work related needs.
- It results from human interactions or social relationship.
- It has no written plans, policies, rules and procedures
- It is guided by customs, conventions culture, group norms, values and belief.
- Its membership is voluntary
- It quickly adapt the environmental change
- It has tendency to resist changes within the group.

Characteristics of informal organization

- The communication is through grapevine or informal channel.
- It coexists with the formal organization
- Its primary focus is person
- It has informal system of reward and punishment
- Its goals are not well-defined and consistent
- Power in informal organization is given by group members rather than delegated by manager
- They do not have well defined tasks; nor they divided and subdivided
- The relationship is interpersonal not impersonal

Advantages of informal organization

- Provides sense of belonging and security to members
- Acts as a safety valve for emotional problems
- Members get help on the job from one another
- Serves as an important channel of communication
- Social control through group norms is possible
- Authority of members can be kept under check
- Reduces need for close supervision by management
- Employees reaction about proposed managerial actions can be known in advance

Disadvantage of informal organizations

- Resistance to change
- Rumor spreading
- Interference on management decision
- Group think philosophy
- Role conflict and sub-optimization

Reasons for emergence or need for informal organization

- To fulfill social security, affiliation, esteem, etc.
- To work in close proximity with group.
- To share view with people with similar social, cultural, economic, etc. composition
- To interacts with others.

Virtual Organization

Virtual Organization (VO) is a group of people or companies that work together remotely to achieve a common goal, using modern Information Technology to communicate and collaborate.

Features:

VOs have no fixed location, and employees can be based anywhere in the world. They use technology to work together in real-time, often across multiple time zones.

Benefits:

VOs can be dynamic and versatile, adapting quickly to change. They can also scale to meet demand.

Challenges

VOs can face challenges such as communication difficulties, isolation, coordination difficulties, and security concerns.

Management

Measuring the effectiveness of a VO can be challenging because of the dynamic nature of the organization and the virtual environment. Business Activity Monitoring (BAM) is an approach that can help monitor a VO's effectiveness by tracking key performance indicators (KPIs) in real-time.

Components of a virtual organization

Each virtual organization is unique, although they often include many of the same components for optimal operations, like a remote workforce and company-specific technology networks. Other components of a virtual organization may include:

- A flat organization structure with less middle management
- Virtual teams
- Loose organizational structure
- Boundaries and expectations
- Power flexibility
- Informal communication

Benefits of a virtual organization

- Lower overhead costs
- Improved employee satisfaction
- Improved efficiency
- Larger hiring market
- Flexible hours
- Improved employee retention
- Access to new markets

Challenges of a virtual organization

- Lack of socialization
- Difficulty developing company culture
- Increased importance of communication
- Potential compliance and security issues

Engineering Management

- Engineering management is a field that combines business management and engineering to help organizations achieve common goals
- It is the application of engineering methods, tools, and techniques to business management systems.
- Engineering management is a career that brings together the technological problem-solving ability of engineering and the organizational, administrative, legal and planning abilities of management in order to oversee the operational performance of complex engineering-driven enterprises.

Importance of Management in Technology Driven Environment

Management is important in a technology-driven environment because it helps businesses:

Adapt to change

- Technology is constantly evolving, and effective technology management helps businesses adapt to new demands and expectations.

Improve efficiency

- Technology management can reduce operational costs and enhance overall efficiency.

Align IT with business goals

- Technology management helps businesses view IT as a strategic asset, and align IT investments with their overall business objectives.

Improve decision-making

- Technology managers can inform strategic decision-making by predicting trends and risks.

Improve customer experience

- Businesses that manage their technology well can deliver better customer experiences.

Improve profitability

- Technology management can transform IT services from a cost to an enabler of profitability.

Some key objectives of IT management include: Boosting operational efficiency, Ensuring security, Managing risks, and Aligning IT with business goals.

Engineering Functions in an Organization

- The primary role of engineers in an organization is to use their technical knowledge to create products, structures, and processes that are valuable to the organization and its customers.
- Here are some other functions of engineering in an organization:
  - Communication
  - Collaboration
  - Team building
  - Product development
  - Decision-making
  - Continuous learning
  - Leadership
  - Organizational skills
  - Project documentation

Product Development

It is the process of creating a new product, improving an existing one, or introducing a product to a new market. It involves a number of steps, including:

- Researching: Conducting market research to understand customer needs
- Ideating: Brainstorming and generating ideas for the product
- Prototyping: Creating a prototype of the product
- Testing: Testing the product and gathering feedback from customers
- Launching: Launching the product to the market in collaboration with marketing

Operation and IT Systems

IT operations are the processes and practices that ensure IT systems and services are available and reliable for business needs. IT operations teams are responsible for a number of tasks, including:

- Help desk operations: Ensuring users can access the tools and systems they need to be efficient
- Device management: Improving the quality and efficiency of IT resources
- Network infrastructure: Ensuring the success of the entire IT infrastructure
- Configuration management: Managing IT assets to maintain productivity
- Performance measurement: Measuring the performance of IT operations

Quality Assurance

It is a systematic process to ensure that products or services meet the expectations of customers and other stakeholders. It involves a range of activities, including:

- Monitoring: Keeping track of customer interactions, such as calls and chats, to ensure quality standards are met
- Evaluating: Assessing whether a team is complying with legal and company regulations, and offering a quality service in a consistent way
- Preventing defects: Designing, documenting, and building quality into processes to prevent defects
- Auditing: Comparing actual conditions with requirements and reporting the results to management

Roles and Responsibilities of an Engineering Manager

An Engineering managers should have strong analytical, communication, and organizational skills. They should also be able to tackle problems on a larger scale and stay up to date on the latest research.

An engineering manager is responsible for many tasks, including:

- Team Management: Hiring, training, and evaluating engineering staff, and ensuring their development
- Technical Leadership: Being the technical lead for technical decisions, and staying up to date on related technologies
- Communication: Ensuring effective communication between team members, other stakeholders, and intra teams

- Leadership: Inspiring and motivating the team, resolving conflicts, and improving cooperation
- Quality assurance: Helping the staff run tests or troubleshoot issues with prototypes
- Progress monitoring :Gauging progress to ensure engineering goals are being met
- Reporting: Preparing progress reports and status updates to share with clients or key stakeholders

Thank You